

EDUCATION

Northeastern University, Boston, MA

Bachelor of Science in Computer Science, Minor in Business Analytics, Minor in Music

Sept 2021 - May 2025

GPA: 3.76 /4.0

TECHNICAL SKILLS

Languages: Java | Python | C | Racket | HTML | CSS | JavaScript | TypeScript | Latex.

Technologies: Git/GitHub, Docker, Dev Containers, React.js, Maven, Gradle, Visual Studio Code, IntelliJ, DataGrip

RELATED WORK EXPERIENCE

Teaching Assistant,

Northeastern (OOD May – Dec 2023 & Jan – April 2025, Logic & Comp Jan -April 2024)

May 2023 – Present

Boston, MA

- Facilitated 14 hours of in-person Office hours in **Java Design Patterns**
- Guided 6 **Java** lab (~600 hours total, 35+ students per lab) every week.
- Mentored 10+ students during lab and assisted 10+ students per week with **GitHub**, design, and debugging issues.
- Evaluated over 200 assignments with individual unique **Java & Racket** code base.

Software Engineer Intern,

John Hancock (.NET, MySQL)

May 2024 - Aug 2024

Boston, MA

- Lead **TLS** deprecation efforts by investigating multiple **.NET** code bases (3+), upwards of 10 branches in **TFS**.
- Coordinated interns on **React.js** to convert an **Angular.js & MySQL CRUD** application to updated **React.js**
- Diagnosed **QA** login authentication issues, categorizing and resolving 91.2% of most errors (discovered to be an **API** fault) & documented in **Atlassian Confluences**.
- Participated in story pointing sessions, alongside daily **scrum** meetings with **Atlassian Jira tickets**.
- Supported automation test team to implement auto regression test suite in **Java** reducing manual testing.

IT Network Intern,

Square Tech (Ubuntu/Linux, Bash Script)

Oct 2024 – Dec 2024

Boston, MA

- Established a Course Management Server (**Moodle**) on **Ubuntu Server**, on site and created documentation on navigation through server for 30+ teachers & students for improved ease of access.
- Curated technical documentation on **LastPass** operation for 50+ employees to strengthen organizational security.
- Maintained 2 server rooms, 3 lab rooms (including desktops), and 30+ laptops throughout **CSTO**.

PROJECTS

Studio Ghibli Viz | Tableau

April 2024

- Curated dataset based on *Howl's Moving Castle* and *Castle in the Sky*.
- Features Dendrogram chart, parameters to switch between movies, song timeline with custom descriptions.
- Published to Tableau Public.

Lyft Dashboard | Tableau

March 2024

- Incorporated two days of Lyft data in dashboard with Lyft Colors theme.
- Displays data via Bullet Graphs, Stacked Bard Charts, Density Map (selectable time) and line graphs.

Content-Delivery Network | Python

Oct 2024 – Dec 2024

- Created a load balancing **DNS** server to 8 proxy **HTTP servers** with storage and query capabilities.
- Optimized data storage with customizable memory & disk limits **LFU cache** management.
- Utilized subprocesses, **SSH**, **SCP** and **SED** to deploy servers into separate remote **Ubuntu 20.04 servers**.

BGP Router | Rust

Feb 2024 - March 2024

- Implemented routing table logic, **selective acknowledgement** and **sliding window** acknowledgment.
- Developed in Rust, Packet, Packet Type, Header, Sender, Receiver structures and implementation utilizing **UDP Sockets** with slow start **Reno Congestion Control**.

Covey Town Chess Game | Website, TypeScript

Feb 2024 - April 2024

- Utilized **Agile** process model and 15-min standups to gauge project progress.
- Remotely hosted websites with **React.js & Node.js** on **Heroku** and **Render**.
- Curated Chess Engine using **TypeScript** with check mate, stalemate and full standard chess interactions.
- Incorporated **Twilio API** for chat/video chat and **Stock Fish API** to create a chess bot with difficulty, tested with **Visual Studio Code Thunder Client**.

Ray Tracer | Java**May 2023 – June 2023**

- Structured **MVC** pattern to isolate Camera View orientation and Model for computation, allowing unilateral upgrade from 2 dimensional (x, y points) to 3 dimensional images (vector calculations).
- Optimized computation by 3 times using **Multi-threading** and Bounding Volume Hierarchy (**BVH**).
- Automated diagnostic data, time stamps, rays rendered and intersection of rays upon completion of an image.

Kanbas | Website, JavaScript, MongoDB**Sept 2023 - Dec 2023**

- Remotely hosted **CRUD** website with **React.js** & **Netlify** and **Node.js** & **Render**.
- Styled the website with **Bootstrap** and **Font Awesome 4** icons with custom overrides to look like Canvas.
- Leveraged **Node.js** to create custom **API** routes for User, Course and Modules, isolating frontend, and backend.
- Utilized **MongoDB Compass** alongside **MongoDB Atlas** to store user data on the cloud and be able to query data.

Snap Map | Website,**24 hours in March 2023**

- Mobilized a minimum viable web application with two other team members, pooling information with 250k locations for low-income communities to find necessary resources related to **SNAP/EBT**.
- Composed an interactive map with **CSV** files (using **Leaflet.js** and **d3.js**) to automatically mark all possible stores within a zip-code (>20), accepting SNAP/EBT and deliver FDA-sanctioned links for further assistance.
- Tested numerous zip code inputs and validate if zip code is valid, if not send alert to user in browser.
- Received the **Best Hack for Community** Award at the White-Hall 2023 Blast to the Past Hackathon.

Titration Viz GUI | Java**June 2022 – July 2022**

- Deployed an MVC **Java/JavaFX** application to support plotting over 12 color coded lines in a CSV file from Logger Pro in the form of a titration curve.
- Delivered and bundled a **Jar file** with **Java** libraries into an **executable (exe) file** to simplify data analysis.
- Utilized **multithreading** (concurrency) to track any percentage when the CSV data is being processed.

ACTIVITIES AND INTERESTS

Northeastern Dragon Dance & Lion Dance Troupe.

- Member Since **Sept 2021** | Lion Dance Choreographer **Sept 2024 - May 2025**

Boston1 Dragon Boat Club + Boston BBB Dragon Boat Club

- Member Since **April 2023**

Other interests: Ukulele, Guitar, Singing, Recording/Producing Songs, Gaming